

KITSOTY AGCM, AB, Canada

WMO: 717970

Lat: 53.3533N

Lon: 110.4161W

Elev: 664

StdP: 93.60

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.5	-28.2	-34.8	0.2	-31.6	-31.3	0.2	-28.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	27.5	17.2	25.8	16.5	24.2	16.1	19.3	24.9	18.2	23.6	17.2	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.4	22.1	16.1	12.4	20.9	15.1	11.6	19.9	58.1	25.1	54.2	23.6	51.0	22.5	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A		DB	-34.6	30.9	3.0	1.8	-36.8	32.2	-38.6	33.3	-40.3	34.3	-42.5	35.6
				WB	-34.7	22.0	3.0	1.3	-36.9	22.9	-38.7	23.6	-40.4	24.4	-42.5	25.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.4	-12.5	-5.3	2.6	10.9	15.0	17.0	16.1	11.4	3.8	-5.2	-12.2
	DBStd	12.48	8.39	8.15	7.18	5.43	4.61	3.19	2.64	3.11	4.70	5.03	6.77	7.58
	HDD10.0	3428	663	629	473	227	45	4	0	1	38	202	457	688
	HDD18.3	5782	922	862	732	472	232	106	58	82	211	452	707	947
	CDD10.0	729	0	0	0	5	73	155	217	189	81	8	0	0
	CDD18.3	42	0	0	0	0	2	7	17	12	3	0	0	0
	CDH23.3	546	0	0	0	2	59	83	172	147	81	2	0	0
	CDH26.7	92	0	0	0	0	6	11	29	26	20	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	403	18	12	16	29	44	73	78	57	33	17	15	19
	PrecMax	593	39	28	42	82	114	143	213	169	92	41	52	32
	PrecMin	216	4	2	3	3	5	12	16	5	2	2	2	5
	PrecStd	91	10	6	8	22	29	31	44	30	25	10	12	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	4.8	11.7	21.6	27.6	28.1	29.8	29.1	29.4	21.4	12.6	4.8
		MCWB	2.7	2.2	5.8	9.7	13.5	17.1	20.3	18.3	16.6	12.7	6.2	1.8
	2%	DB	3.4	2.5	7.9	17.6	25.1	25.7	26.9	26.8	25.6	16.5	8.0	2.7
		MCWB	1.4	0.6	3.9	8.1	12.8	15.7	18.7	17.5	15.1	9.8	4.3	0.6
	5%	DB	2.0	0.7	5.2	14.2	22.3	23.7	25.2	24.8	22.7	13.8	4.9	0.4
		MCWB	0.4	-0.6	2.5	6.6	11.6	15.1	18.0	17.3	14.0	8.3	2.5	-1.5
	10%	DB	0.1	-0.8	3.2	11.2	19.9	21.8	23.4	23.1	19.7	11.2	2.7	-2.3
		MCWB	-1.2	-1.9	1.2	5.0	10.6	14.1	17.0	16.4	12.6	6.9	0.9	-3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.5	6.7	10.3	15.0	18.9	21.9	20.6	17.5	13.1	7.4	2.2
		MCDB	5.0	4.6	11.3	20.9	23.6	25.0	27.1	26.5	26.7	20.5	11.5	4.3
	2%	WB	1.6	0.9	4.1	8.4	13.5	17.2	19.8	19.0	15.8	10.8	4.7	0.7
		MCDB	3.2	2.3	7.4	16.5	22.0	23.2	25.3	24.5	24.2	15.2	7.7	2.8
	5%	WB	0.4	-0.4	2.6	6.9	12.5	16.2	18.7	17.9	14.4	8.9	2.6	-1.4
		MCDB	1.9	0.6	4.9	13.4	20.6	21.4	24.0	23.2	21.4	13.0	4.6	0.2
	10%	WB	-1.3	-1.9	1.3	5.5	11.4	15.1	17.6	16.9	13.1	7.2	1.1	-3.4
		MCDB	0.0	-0.7	3.1	10.5	18.9	20.4	22.4	21.9	18.9	11.1	2.6	-2.2
(35) Mean Daily Temperature Range	MDBR		8.8	9.3	8.8	10.5	12.5	10.7	10.9	11.3	11.9	9.6	7.6	7.5
	5% DB	MCDBR	9.0	9.9	10.6	15.5	15.6	13.7	13.6	14.4	16.8	13.8	10.5	9.6
		MCWBR	7.5	8.4	7.2	8.1	6.6	6.0	6.9	7.1	7.7	7.8	7.1	7.9
	5% WB	MCDBR	8.8	9.6	10.0	14.7	14.0	11.5	12.4	12.7	15.7	12.7	9.6	9.6
		MCWBR	7.5	8.4	7.0	8.0	6.4	5.9	6.9	6.8	7.7	7.6	6.8	8.2
(40) Clear-Sky Solar Irradiance	taub		0.242	0.291	0.281	0.326	0.347	0.366	0.362	0.372	0.309	0.274	0.248	0.217
	taud		2.333	2.195	2.394	2.377	2.365	2.339	2.377	2.326	2.484	2.552	2.452	2.361
	Ebn at Noon		766	804	917	908	901	884	881	848	874	846	763	742
	Edh at Noon		55	89	91	106	113	118	112	111	83	62	49	44
(44) All-Sky Solar Radiation	RadAvg		0.99	2.05	3.55	4.70	5.82	5.95	6.16	5.02	3.54	2.05	0.99	0.74
	RadStd		0.10	0.15	0.28	0.41	0.41	0.43	0.34	0.38	0.39	0.29	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page